
Parameter Scope Matters: MNE-L2 for Reliable Low-Latency ANN-to-SNN Conversion

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Abstract

Spiking neural networks (SNNs) use sparse, discrete spike events to support event-driven and potentially energy-efficient inference on neuromorphic hardware. ANN-to-SNN conversion offers a practical route to deep SNNs by reusing well-trained ANN parameters, thereby avoiding the difficulty of training deep SNNs directly. However, thresholded spike representations can be fragile under noise. To investigate how source training shapes this robustness, we study the parameter scope of regularization and identify a failure: applying L2 to batch-normalization (BN) affine parameters can make converted SNNs substantially less robust than applying it only to weights. We analyze this failure through local spike-count boundary crossings and propose margin-normalized effective L2 (MNE-L2), which rescales the L2 penalty on source weights by BN-folded amplification relative to each layer’s conversion margin. Across five-seed CIFAR experiments, MNE-L2 improves high-noise accuracy over all-parameter L2 by up to 62.06 percentage points while reducing clean firing activity and estimated arithmetic energy. These findings expose regularization scope as an important training choice and demonstrate that MNE-L2 improves the reliability–efficiency trade-off for low-latency SNN inference.

1 Introduction

Spiking neural networks (SNNs) communicate through discrete spike events rather than dense activations, enabling sparse, event-driven computation and potentially energy-efficient inference on neuromorphic hardware [1]. Realizing these advantages with deep models, however, requires accurate inference within a short simulation horizon. High-accuracy deep SNNs can be obtained either by direct training or by converting artificial neural networks (ANNs). ANN-to-SNN conversion is particularly practical because it reuses well-trained ANN parameters and avoids the difficulty of optimizing deep SNNs directly [2, 3]. Converted SNNs often require many simulation steps to approximate ANN activations. Reducing the timestep budget lowers latency and event cost but increases finite-time conversion error [4]. Recent conversion and calibration methods have substantially improved low-timestep accuracy [5, 4], making reliability under deployment noise increasingly important.

Because SNN inference is built from thresholded spike events, perturbations can abruptly alter spike counts and propagate through later layers. Prior work has studied noise sensitivity in trained SNNs and adversarial robustness after conversion [6, 7]. Most low-latency conversion methods, however, focus on reducing clean conversion error through activation design, calibration, or membrane-potential correction. How the source ANN’s training choices shape post-conversion robustness has received much less attention.

To fill this gap, we start with a controlled ablation of L2 regularization, a standard source-ANN training choice whose parameter scope is often left implicit. In networks with batch normalization (BN) [8], an L2 penalty can cover weights alone or also the BN affine parameters. Figure 1 compares weights-only L2 with penalties that additionally include BN- β , BN- γ , or both. Including BN- γ causes a sharp accuracy collapse even when moderate Gaussian noise ($\sigma = 1$) is applied independently at each SNN timestep, whereas weights-only L2 and weights plus BN- β remain stable. This scope-dependent failure motivates our stability analysis.

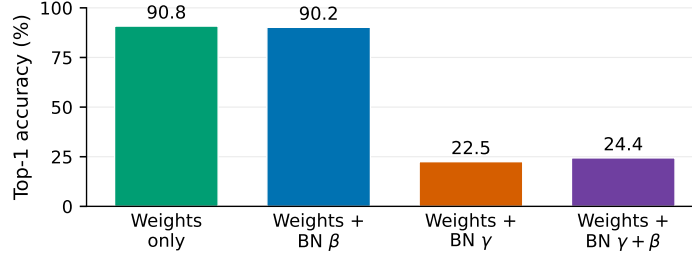


Figure 1: CIFAR-10 VGG16 parameter-scope ablation with Gaussian noise applied independently at each SNN timestep. At $\sigma = 1$, including BN- γ , with or without BN- β , causes a sharp accuracy collapse.

To explain this scope-dependent failure, we analyze how additive perturbations cross local spike-count boundaries as they propagate through a converted network. Local risk is governed by effective noise relative to the nearest transition margin: downstream BN-folded weights determine perturbation amplification, while the following integrate-and-fire (IF) threshold sets the nominal conversion margin. Guided by this analysis, we propose margin-normalized effective L2 (MNE-L2). It rescales L2 updates on convolutional and linear weights so that layers with larger folded amplification relative to their margin receive stronger regularization, while BN affine parameters and IF thresholds receive no gradient from the regularization term.

Our contributions are:

- We identify an underexplored L2 parameter-scope failure and isolate BN- γ as the affine term that distinguishes the unstable settings in a controlled four-way ablation.
- We derive a local boundary-crossing view and layerwise margin-to-noise diagnostics that motivate margin-scaled regularization of source-ANN weights.
- We show across CIFAR VGG13/16/19 and ImageNet ResNet-18 that MNE-L2 substantially improves high-noise accuracy over all-parameter L2 and, on CIFAR, reduces firing density and estimated arithmetic energy.

2 Related Work

Low-latency ANN-to-SNN conversion. Early ANN-to-SNN conversion methods rely on weight normalization, threshold balancing, and activation normalization [9, 2]. At short simulation horizons, finite-time mismatch has been reduced through optimized potential initialization [10], parameter calibration [11], residual-membrane-potential and offset-spike correction [12, 13], mitigation of temporal misalignment with probabilistic spiking neurons [14], and quantization-aware ANN training [5, 4]. SpikeConverter reduces the activation-to-spike representation gap through specialized conversion dynamics [15], while more recent methods learn explicit conversion-error compensation or bridge quantized activations and spiking neurons [16, 17]. Collectively, these methods improve clean conversion accuracy or latency by modifying activation design, calibration, or conversion dynamics. In contrast, we retain QCFS as a fixed conversion backbone and study how source-weight regularization affects post-conversion robustness.

Regularization scope in normalized networks. Normalization can make preceding weights scale-invariant, causing L2 regularization to act through parameter scale and the effective learning rate rather than solely as a conventional capacity penalty [18]. Practical CNN training recipes often

restrict weight decay to convolutional and fully connected weights while excluding biases and BN affine parameters [19]. More generally, whether BN- γ should be regularized depends on its architectural role, and practices vary across libraries and implementations [20]. These studies concern ANN optimization and clean-task performance. We instead study how regularization scope shapes internal-noise stability after conversion; MNE-L2 scales source-weight regularization by BN-folded amplification relative to the boundary margin of the following IF activation.

Stability and noise sensitivity. Perturbation stability has been studied through Lyapunov bounds for ANNs [21] and membrane-potential dynamics in trained SNNs [6]. Related work has also analyzed average input-noise stability and Fourier-spectrum concentration in wide LIF-SNN classifiers [22], as well as classifier-level adversarial robustness after conversion [7]. We instead derive a neuron-level upper bound on the probability that internal Gaussian noise crosses a spike-count boundary in a converted SNN. This local analysis motivates MNE-L2 and corresponding diagnostics for spike-count stability, rather than worst-case classifier robustness.

3 Method

3.1 Low-latency QCFS conversion

We first describe the quantization clip-floor-shift (QCFS) conversion used in our experiments and then derive the proposed regularizer. For activation x , threshold λ , and quantization level L , the source ANN uses [4]

$$q_L(x; \lambda) = \frac{\lambda}{L} \left\lfloor L \operatorname{clip}\left(\frac{x}{\lambda}, 0, 1\right) + \frac{1}{2} \right\rfloor. \quad (1)$$

At inference, an integrate-and-fire (IF) layer emits a length- T spike sequence whose rate represents the quantized activation. Unless stated otherwise, we use $L = T = 16$ and a rate-uniform schedule for the VGG experiments. Additional depth-transfer and compact low-latency controls are reported only in the appendix.

3.2 Local noise-to-margin analysis

Let z be the clean output of an IF layer and $\tilde{z} = z + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$. Let $\mathcal{B}_l$ contain the QCFS boundaries of layer l and $d_l(u) = \min_{b \in \mathcal{B}_l} |u - b|$. Under a local additive model $\delta_l \sim \mathcal{N}(0, \sigma_{\text{eff},l}^2)$, changing the quantization bin requires crossing a boundary, so

$$\Pr[q_L(u + \delta_l) \neq q_L(u) \mid u] \leq 2 \mathcal{Q}\left(\frac{d_l(u)}{\sigma_{\text{eff},l}}\right), \quad (2)$$

where $\mathcal{Q}$ is the Gaussian upper-tail function. We estimate $\sigma_{\text{eff},l} = \sqrt{\mathbb{E}[(z_l - \tilde{z}_l)^2]}$, summarize $d_l(u)$ by its median $d_{50,l}$, and report

$$\rho_l = \frac{d_{50,l}}{\sigma_{\text{eff},l} + \varepsilon}, \quad P_{\text{cross},l} = \Pr[q_L(z_l) \neq q_L(\tilde{z}_l)]. \quad (3)$$

Small ρ_l and large $P_{\text{cross},l}$ identify layers where noise is likely to change discrete spike counts. In an idealized independent aggregation model, σ_{eff}^2 is upper-bounded by $O(1/(CT))$, while the nominal QCFS half-cell margin decreases as $1/L$; hence a nominal stability proxy scales as $\sqrt{CT}/L$ (Appendix A). This motivates evaluating precision, width, and timestep jointly rather than treating L as an isolated accuracy knob. It also suggests controlling the BN-folded amplification that contributes to $\sigma_{\text{eff},l}$ relative to the following QCFS margin.

3.3 Margin-aware weight regularization

Guided by the preceding analysis, for each convolution or linear layer followed by BN and an IF activation, we fold the BN scale into the matched weight while stopping the regularization gradient through BN and the IF threshold:

$$\widetilde{W}_{l,o} = \operatorname{sg}\left(\frac{\gamma_{l,o}}{\sqrt{v_{l,o} + \epsilon_{\text{BN}}}}\right) W_{l,o}, \quad M_{\text{eff},l} = \frac{1}{C_l} \sum_o \|\widetilde{W}_{l,o}\|_F^2. \quad (4)$$

Margin-normalized effective L2 is

$$\mathcal{R}_{\text{MNE}} = \sum_l \frac{L_l^2 M_{\text{eff},l}}{\text{sg}(\lambda_l)^2 + \varepsilon}, \quad (5)$$

where sg denotes stop-gradient. Layers without a following IF threshold are excluded. Because the nominal QCFS half-cell margin is $\lambda_l/(2L_l)$, the factor L_l^2/λ_l^2 is proportional to its inverse square. MNE-L2 does not optimize the margin directly; it uses the current margin and BN-folded amplification to scale the L2 gradient on W_l , applying stronger weight regularization where the amplification-to-margin ratio is larger. BN- γ , BN- β , and λ_l receive no gradient from this term. Thus, all-parameter, weights-only, and selective L2 penalties are different interventions even when their coefficients are numerically similar.

4 Experiments

4.1 Setup

We evaluate QCFS VGG16 on CIFAR-10/100 [23] and QCFS ResNet-18 on ImageNet [24, 25]. Unless noted, $L = T = 16$, with independent Gaussian noise injected after the input IF at each timestep. L2-all regularizes all trainable parameters; L2-wo restricts the same penalty to convolutional and linear weights, and L1-wo uses the same weights-only scope. CIFAR curves report mean $\pm$ sample standard deviation over independently trained seeds 40–44; ImageNet uses one seed-42 checkpoint. The CIFAR-10 scope ablation compares weights only, weights plus BN- β , weights plus BN- γ , and both affine terms using seed 42 and one stored perturbation stream. Hyperparameters are fixed before noise evaluation, and configurations are not clean-accuracy matched. This synthetic single-site stress test is neither an adversarial attack nor a device measurement; full protocols and additional controls appear in the appendix.

4.2 BN gain distinguishes the unstable scope settings

Figure 1 gives the $\sigma = 1$ snapshot, while Figure 2 shows the full scope sweep. Weights-only L2 and weights plus BN- β retain 75.63% and 70.65% at $\sigma = 3$; both settings containing BN- γ reach chance by $\sigma \approx 1.5$. Their near-overlap implicates gain, rather than bias, as the principal scope difference in this controlled run.

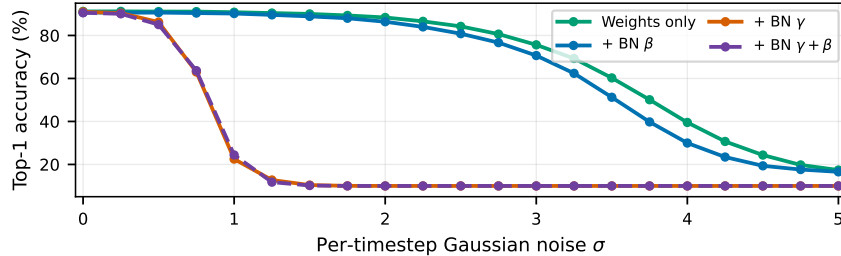


Figure 2: Full CIFAR-10 VGG16 seed-42 parameter-scope sweep. Each point uses the same stored perturbation stream for the four scope settings.

Late-block diagnostics are consistent with this result: all-parameter L2 and weights plus BN- γ reach $\rho_l \approx 0.05$ – 0.11 and $P_{\text{cross},l} \approx 0.62$ – 0.74 , whereas MNE-L2 and weights-only L2 retain larger margins and mostly stay below 0.15 crossings (Appendix B). The one-seed intervention does not establish a universal BN mechanism, but shows why “L2” should report its parameter scope.

4.3 MNE-L2 across CIFAR and ImageNet

Figure 3(a,b) extends the stress test to five checkpoints per CIFAR dataset. At $\sigma = 5$, MNE-L2 reaches 72.06% on CIFAR-10 and 41.14% on CIFAR-100, versus 39.40% and 17.91% for weights-only L1; all-parameter L2 collapses near $\sigma = 1$. The ordering persists for VGG13/19 (Appendix C.2).

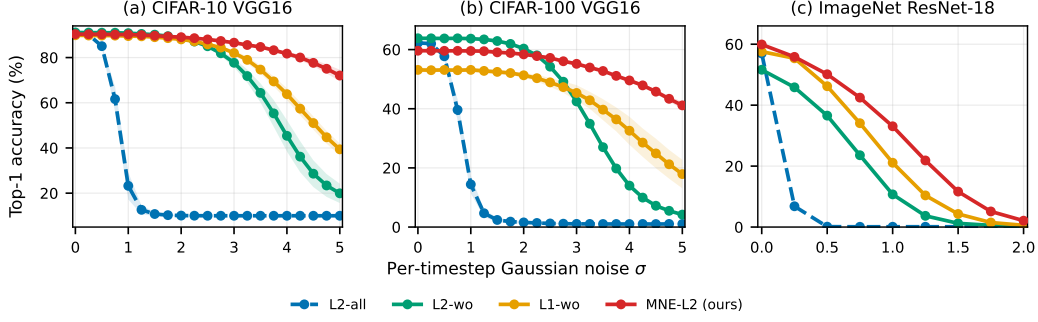


Figure 3: Per-timestep Gaussian-noise sweeps at $L = T = 16$. (a,b) CIFAR VGG16 curves are five-seed means; bands show sample standard deviations across seeds 40–44. (c) ImageNet ResNet-18 uses one seed-42 checkpoint and therefore has no uncertainty band. MNE-L2 is highlighted in red and all data points use circular markers.

Table 1: Five-seed CIFAR VGG16 robustness–activity summary at $T = 16$ (mean $\pm$ sample standard deviation; 10,000 test images per seed). Accuracy is measured under per-timestep Gaussian noise at $\sigma = 5$; firing density and arithmetic energy use clean inference. The best value in each dataset and metric is bold, and the second-best is underlined. Clean accuracy and full activity distributions appear in Appendix C.1.

Dataset	Method	Clean firing density	Clean energy (mJ)	Acc. at $\sigma = 5$ (%)
CIFAR-10	L2-all	0.105 $\pm$ 0.003	0.585 $\pm$ 0.006	10.00 $\pm$ 0.00
	L2-wo	0.102 $\pm$ 0.001	0.505 $\pm$ 0.003	19.92 $\pm$ 4.29
	L1-wo	0.077 $\pm$ 0.001	0.402 $\pm$ 0.005	39.40 $\pm$ 2.60
	MNE-L2 (ours)	<u>0.083 $\pm$ 0.001</u>	<u>0.472 $\pm$ 0.003</u>	72.06 $\pm$ 2.57
CIFAR-100	L2-all	0.106 $\pm$ 0.003	0.558 $\pm$ 0.008	1.05 $\pm$ 0.06
	L2-wo	0.105 $\pm$ 0.001	0.522 $\pm$ 0.002	4.21 $\pm$ 0.24
	L1-wo	0.084 $\pm$ 0.002	0.429 $\pm$ 0.005	17.91 $\pm$ 4.84
	MNE-L2 (ours)	<u>0.090 $\pm$ 0.001</u>	<u>0.494 $\pm$ 0.004</u>	41.14 $\pm$ 1.23

On the single ImageNet checkpoint, MNE-L2 retains 33.07% at $\sigma = 1$, versus 21.11% for L1-wo, and remains highest at $\sigma = 2$ (Figure 3(c)).

4.4 Clean sparsity, robustness, and estimated energy

Table 1 compares all four regularizers at high noise. On CIFAR-10, MNE-L2 lowers clean firing from 0.105 to 0.083 and energy from 0.585 to 0.472 mJ relative to L2-all, while retaining 72.06% rather than 10.00% accuracy at $\sigma = 5$. The same robustness ordering holds on CIFAR-100, where MNE-L2 reaches 41.14% and L2-all reaches 1.05%. L1-wo is the sparsest and least energy-intensive method, but Appendix C.1 shows that it trails MNE-L2 by 6.49 clean CIFAR-100 points.

Following Horowitz [26], estimated energy charges repeated first-layer MACs at 4.6 pJ and clean event-triggered ACs thereafter at 0.9 pJ (Appendix Eq. 9). Noisy firing is excluded because injected events would confound deployment cost. The estimates omit memory, neuron updates, control, interconnect, and leakage, and are not device measurements.

5 Implications, Limitations, and Conclusion

Regularizer names should specify parameter scope because including BN- γ changes the observed robustness ranking. MNE-L2 uses folded BN scales and IF margins to regularize only convolutional and linear weights, yielding lower activity than L2-all. Limitations are a one-seed scope ablation, one ImageNet checkpoint, synthetic single-site noise, unmatched clean accuracy, arithmetic-only energy, and VGG-only five-seed CIFAR results. These preclude universal dominance, but MNE-L2 better preserves post-IF representations than L2-all in the tested settings.

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A A Local Precision–Width–Time Model

Consider an idealized normalized aggregation of independent perturbations,

$$\delta_{C,T} = \frac{1}{CT} \sum_{t=1}^T \sum_{i=1}^C w_i \epsilon_{i,t}, \quad \epsilon_{i,t} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2). \quad (6)$$

If $C^{-1} \sum_i w_i^2 \leq M$, then

$$\text{Var}(\delta_{C,T}) = \frac{\sigma^2}{C^2 T} \sum_i w_i^2 \leq \frac{\sigma^2 M}{CT}. \quad (7)$$

For an unclipped QCFS cell, the nominal half-cell margin is $h_L = \lambda/(2L)$. Combining this value with the variance bound gives the nominal proxy

$$\rho_{\text{nom}} = \frac{h_L}{\sigma_{\text{eff}}} \geq \frac{\lambda \sqrt{CT}}{2L \sigma \sqrt{M}}. \quad (8)$$

Equation (2) follows because a changed quantization code requires $|\delta_l| \geq d_l(u)$ and a zero-mean Gaussian satisfies $\Pr(|\delta_l| \geq d) = 2Q(d/\sigma_{\text{eff},l})$. This local model isolates the competing directions of C , T , and L ; it is not a claim that all network perturbations are independent or Gaussian, nor a proof that the empirical accuracy gap must be monotone.

Table 2: Five-seed clean MNIST conversion at $T = 4$ (mean $\pm$ standard deviation, no injected noise). Δ is $L=16$ minus $L=4$ SNN accuracy.

Channels	SNN, $L = 4$	SNN, $L = 16$	Δ
2/4	96.14 ± 0.91	87.80 ± 6.94	-8.34
4/8	98.00 ± 0.26	97.01 ± 0.60	-0.99
8/16	98.67 ± 0.13	98.22 ± 0.37	-0.44
16/32	98.67 ± 0.12	98.42 ± 0.21	-0.25

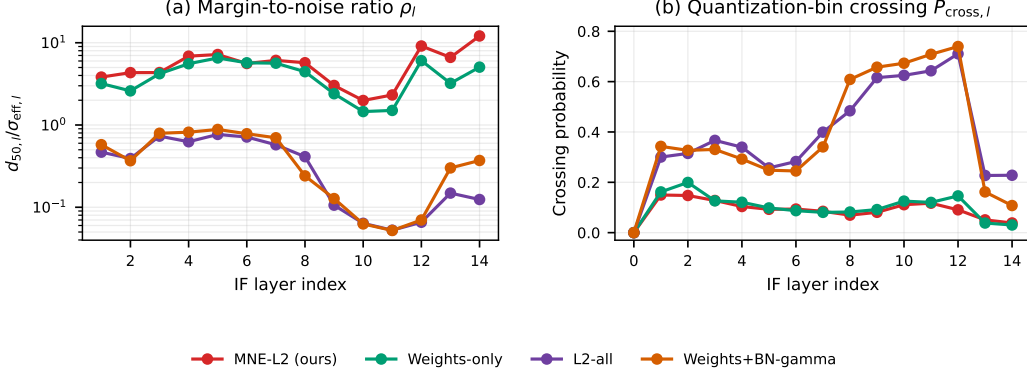


Figure 4: Layerwise CIFAR-10 VGG16 diagnostics at fixed $\sigma = 1$. All-parameter L2 and weights+BN- γ checkpoints develop small margin-to-noise ratios and high bin-crossing probability in later blocks.

253 A.1 Compact precision-width control

254 To separate conversion error from injected-noise calibration, we train weight-decayed MNIST CNNs
 255 with channel pairs (2, 4), (4, 8), (8, 16), and (16, 32) at $L \in \{4, 16\}$. Each setting uses five training
 256 seeds and is converted at $T = 4$ with no injected noise. Raising L from 4 to 16 improves every source
 257 ANN by 0.06–0.29 points, yet reduces $T = 4$ SNN accuracy by 8.34 points for the 2/4-channel
 258 model. This loss contracts to 0.25 points at 16/32 channels (Table 2).

259 B Layerwise Diagnostics

260 Figure 4 reports the two diagnostics from Eq. (3). Layer 0 is the perturbed input IF; its effective-noise
 261 denominator is zero by construction for the downstream-propagation statistic and is omitted from the
 262 left panel. The right panel directly measures whether clean and noisy IF outputs differ.

263 C Additional Experimental Details

264 The CIFAR VGG16 source ANNs are trained for 300 epochs with batch size 128 and seeds 40–44
 265 where multi-seed results are reported. Training uses SGD with momentum 0.9, initial learning rate 0.1,
 266 and cosine annealing. Random crops with four-pixel padding, horizontal flips, CIFAR AutoAugment,
 267 and Cutout are used during training; evaluation uses only dataset-specific normalization. MNE-L2
 268 uses coefficient 10^{-4} with detached BN affine/statistical factors and detached IF thresholds. All-
 269 parameter L2 uses weight decay 5×10^{-4} , weights-only L1 uses 10^{-5} , and the selective L2 scope
 270 controls use coefficient 2.5×10^{-4} . The scope comparison reuses CIFAR-10 seed-42 checkpoints
 271 and the same evaluation protocol at every severity.

272 The ImageNet ResNet-18 source ANNs are trained at $T = 0$ for 90 epochs with seed 42, batch size
 273 256, SGD momentum 0.9, initial learning rate 0.1, and cosine annealing. Training applies a random
 274 224×224 resized crop and horizontal flip; validation resizes to 256 pixels and takes a 224×224
 275 center crop. Both splits use standard ImageNet normalization. The highest clean validation-accuracy
 276 checkpoint is converted with $L = T = 16$ and evaluated on the 50,000-image validation split.

Table 3: Top-1 accuracy under per-timestep Gaussian noise at $\sigma = 1$ (mean $\pm$ sample standard deviation across seeds 40–44).

Dataset	Method	$T = 4$	$T = 8$	$T = 16$
CIFAR-10	L2-all	10.10 ± 0.15	11.17 ± 1.18	23.30 ± 6.46
	MNE-L2 (ours)	88.97 ± 0.21	89.76 ± 0.18	90.09 ± 0.10
CIFAR-100	L2-all	1.56 ± 0.40	2.88 ± 0.83	14.64 ± 3.16
	MNE-L2 (ours)	58.77 ± 0.28	59.27 ± 0.11	59.48 ± 0.23

Table 4: Full clean SNN top-1 accuracy (mean $\pm$ sample standard deviation across seeds 40–44).

Dataset	Method	$T = 4$	$T = 8$	$T = 16$
CIFAR-10	L2-all	90.29 ± 0.23	90.52 ± 0.14	90.59 ± 0.13
	L2-wo	90.57 ± 0.18	90.96 ± 0.06	91.10 ± 0.07
	L1-wo	89.09 ± 0.25	89.70 ± 0.30	89.87 ± 0.20
	MNE-L2 (ours)	89.93 ± 0.16	90.31 ± 0.17	90.38 ± 0.14
CIFAR-100	L2-all	61.83 ± 0.47	62.22 ± 0.49	62.27 ± 0.37
	L2-wo	62.68 ± 0.54	63.57 ± 0.31	63.83 ± 0.26
	L1-wo	52.37 ± 0.95	52.89 ± 0.71	53.13 ± 0.73
	MNE-L2 (ours)	59.24 ± 0.21	59.62 ± 0.31	59.62 ± 0.22

277 All-parameter and weights-only L2 use coefficient 10^{-4} , MNE-L2 uses 10^{-4} , and weights-only
 278 L1 uses 10^{-5} . No coefficient sweep or clean-accuracy matching was performed for the reported
 279 ImageNet comparison, and coefficients were not selected from the noisy evaluation curves.

280 For the CIFAR arithmetic estimate, dense ANN energy and event-driven SNN energy are

$$E_{\text{ANN}} = N_{\text{MAC}} E_{\text{MAC}}, \quad E_{\text{SNN}} = T N_{\text{MAC}}^{(1)} E_{\text{MAC}} + N_{\text{AC}}^{(>1)} E_{\text{AC}}, \quad (9)$$

281 with $E_{\text{MAC}} = 4.6$ pJ and $E_{\text{AC}} = 0.9$ pJ. The first convolution receives analog pixels at every
 282 timestep; all later convolutional and linear layers use measured presynaptic events. Only clean
 283 ($\sigma = 0$) activity is used in this estimate.

284 C.1 Full CIFAR robustness and energy statistics

285 Table 3 reports the full $\sigma = 1$ timestep sweep for the main comparison. Tables 4 and 5 report
 286 clean accuracy and energy at every timestep; Table 6 adds per-sample activity distributions and the
 287 L2-wo/L1 controls at $T = 16$. The main-paper Table 1 reports the four-method $\sigma = 5$ comparison.
 288 For energy accounting, we count dense operations in the first convolution at every timestep and
 289 event-triggered ACs in all later convolutional and linear layers. Every table uses all 10,000 examples
 290 in each test set; uncertainty reflects variation across five independently trained checkpoints, not
 291 measurement noise from physical hardware.

Table 5: Full clean CIFAR VGG16 arithmetic-energy estimates (mean $\pm$ sample standard deviation across seeds 40–44). ANN energy is 1.5277 mJ on CIFAR-10 and 1.5294 mJ on CIFAR-100. “Ratio” is ANN energy divided by estimated SNN energy.

Dataset	Method	$T = 4$		$T = 8$		$T = 16$	
		Energy (mJ)	Ratio	Energy (mJ)	Ratio	Energy (mJ)	Ratio
CIFAR-10	L2-all	0.1421 $\pm$ 0.0015	10.75 $\pm$ 0.11	0.2877 $\pm$ 0.0030	5.31 $\pm$ 0.05	0.5848 $\pm$ 0.0060	2.61 $\pm$ 0.03
	L2-wo	0.1227 $\pm$ 0.0008	12.45 $\pm$ 0.08	0.2483 $\pm$ 0.0016	6.15 $\pm$ 0.04	0.5050 $\pm$ 0.0033	3.03 $\pm$ 0.02
	L1-wo	0.0975 $\pm$ 0.0010	15.67 $\pm$ 0.17	0.1969 $\pm$ 0.0022	7.76 $\pm$ 0.08	0.4019 $\pm$ 0.0046	3.80 $\pm$ 0.04
	MNE-L2 (ours)	0.1141 $\pm$ 0.0005	13.39 $\pm$ 0.06	0.2312 $\pm$ 0.0012	6.61 $\pm$ 0.03	0.4716 $\pm$ 0.0026	3.24 $\pm$ 0.02
CIFAR-100	L2-all	0.1354 $\pm$ 0.0020	11.30 $\pm$ 0.16	0.2742 $\pm$ 0.0039	5.58 $\pm$ 0.08	0.5577 $\pm$ 0.0078	2.74 $\pm$ 0.04
	L2-wo	0.1271 $\pm$ 0.0005	12.03 $\pm$ 0.04	0.2569 $\pm$ 0.0010	5.95 $\pm$ 0.02	0.5225 $\pm$ 0.0021	2.93 $\pm$ 0.01
	L1-wo	0.1043 $\pm$ 0.0011	14.66 $\pm$ 0.16	0.2109 $\pm$ 0.0025	7.25 $\pm$ 0.09	0.4294 $\pm$ 0.0052	3.56 $\pm$ 0.04
	MNE-L2 (ours)	0.1197 $\pm$ 0.0009	12.77 $\pm$ 0.10	0.2424 $\pm$ 0.0019	6.31 $\pm$ 0.05	0.4942 $\pm$ 0.0037	3.09 $\pm$ 0.02

Table 6: Clean activity distribution, clean arithmetic energy, and $\sigma = 1$ robustness at $T = 16$. Within each seed, firing density is the fraction of nonzero entries across all IF outputs and is summarized over 10,000 samples by its mean, median, and 95th percentile; entries report the mean $\pm$ sample standard deviation of these statistics across seeds 40–44. Energy uses clean mean activity only.

Dataset	Method	Clean acc. (%)	Fire mean	Fire median	Fire p95	Energy (mJ)	Acc. at $\sigma = 1$ (%)
CIFAR-10	L2-all	90.59 $\pm$ 0.13	0.105 $\pm$ 0.003	0.105 $\pm$ 0.003	0.120 $\pm$ 0.004	0.585 $\pm$ 0.006	23.30 $\pm$ 6.46
	MNE-L2 (ours)	90.38 $\pm$ 0.14	0.083 $\pm$ 0.001	0.083 $\pm$ 0.001	0.099 $\pm$ 0.002	0.472 $\pm$ 0.003	90.09 $\pm$ 0.10
	L2-wo	91.10 $\pm$ 0.07	0.102 $\pm$ 0.001	0.103 $\pm$ 0.001	0.118 $\pm$ 0.001	0.505 $\pm$ 0.003	90.82 $\pm$ 0.11
	L1-wo	89.87 $\pm$ 0.20	0.077 $\pm$ 0.001	0.077 $\pm$ 0.001	0.090 $\pm$ 0.002	0.402 $\pm$ 0.005	89.60 $\pm$ 0.16
CIFAR-100	L2-all	62.27 $\pm$ 0.37	0.106 $\pm$ 0.003	0.106 $\pm$ 0.003	0.121 $\pm$ 0.004	0.558 $\pm$ 0.008	14.64 $\pm$ 3.16
	MNE-L2 (ours)	59.62 $\pm$ 0.22	0.090 $\pm$ 0.001	0.090 $\pm$ 0.001	0.107 $\pm$ 0.002	0.494 $\pm$ 0.004	59.48 $\pm$ 0.23
	L2-wo	63.83 $\pm$ 0.26	0.105 $\pm$ 0.001	0.106 $\pm$ 0.001	0.120 $\pm$ 0.001	0.522 $\pm$ 0.002	63.79 $\pm$ 0.25
	L1-wo	53.13 $\pm$ 0.73	0.084 $\pm$ 0.002	0.085 $\pm$ 0.002	0.101 $\pm$ 0.002	0.429 $\pm$ 0.005	53.08 $\pm$ 0.76

292 C.2 Depth transfer across VGG13 and VGG19

293 We repeat the per-timestep Gaussian-noise sweep at $L = T = 16$ using VGG13 and VGG19 on
 294 CIFAR-10 and CIFAR-100. Each curve reports the mean and sample standard deviation across
 295 independently trained seeds 40–44. In addition to the four main regularizers, this control includes
 296 Calibrated MNE with the fixed $\alpha = 0.1$ configuration; α is a prescribed training schedule rather than
 297 a gradient-triggered or adaptive choice.

298 Across VGG13/16/19, MNE-L2 attains 74.28/72.06/73.23% at $\sigma = 5$ on CIFAR-10 and
 299 44.01/41.14/37.60% on CIFAR-100. Calibrated MNE closely follows L2-wo despite matching
 300 or slightly exceeding its clean accuracy, while L1-wo incurs a substantial CIFAR-100 clean-accuracy
 301 cost that grows from VGG13 to VGG19. Thus the high-noise MNE-L2 advantage transfers across
 302 the three completed depths, although the CIFAR-100 endpoint declines with depth.

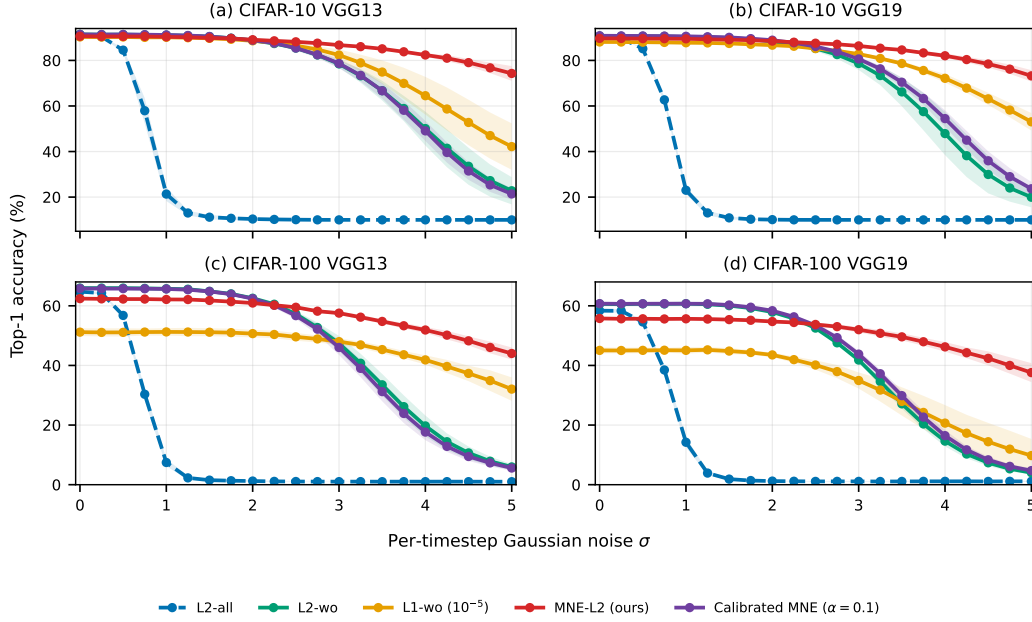


Figure 5: Depth-transfer robustness at $L = T = 16$ under Gaussian noise applied independently at each SNN timestep. Curves are five-seed means and bands show sample standard deviations. MNE-L2 is highlighted in red; Calibrated MNE is included only as an appendix control.

Table 7: Additional-depth accuracy endpoints (mean $\pm$ sample standard deviation across seeds 40–44). The best value within each architecture, dataset, and noise condition is bold.

Architecture	Method	CIFAR-10		CIFAR-100	
		$\sigma = 0$	$\sigma = 5$	$\sigma = 0$	$\sigma = 5$
VGG13	L2-all	90.93 $\pm$ 0.14	9.99 $\pm$ 0.01	64.73 $\pm$ 0.19	1.01 $\pm$ 0.01
	L2-wo	91.44 $\pm$ 0.14	22.78 $\pm$ 5.99	65.94 $\pm$ 0.15	5.96 $\pm$ 0.96
	L1-wo	90.30 $\pm$ 0.14	42.17 $\pm$ 10.20	51.16 $\pm$ 1.31	32.06 $\pm$ 3.92
	MNE-L2 (ours)	90.61 $\pm$ 0.19	74.28 $\pm$ 3.25	62.41 $\pm$ 0.15	44.01 $\pm$ 2.02
	Cal. MNE	91.41 $\pm$ 0.13	21.38 $\pm$ 2.58	65.82 $\pm$ 0.22	5.59 $\pm$ 0.92
VGG19	L2-all	90.19 $\pm$ 0.12	10.00 $\pm$ 0.00	58.37 $\pm$ 0.64	1.13 $\pm$ 0.18
	L2-wo	90.69 $\pm$ 0.18	19.95 $\pm$ 4.44	60.62 $\pm$ 0.32	4.03 $\pm$ 0.65
	L1-wo	88.09 $\pm$ 0.25	53.04 $\pm$ 3.85	45.04 $\pm$ 1.01	9.77 $\pm$ 5.59
	MNE-L2 (ours)	89.67 $\pm$ 0.06	73.23 $\pm$ 2.50	55.75 $\pm$ 0.43	37.60 $\pm$ 3.19
	Cal. MNE	90.84 $\pm$ 0.03	23.66 $\pm$ 3.05	60.73 $\pm$ 0.22	4.73 $\pm$ 0.76

303 C.3 Compact MNIST-family controls

304 We use MNIST [27] and Fashion-MNIST [28] with a 2,081-parameter CNN: two 3×3 convolution-
 305 BN-IF blocks with 2 and 4 channels, 2×2 max pooling, and a linear classifier. Source ANNs are
 306 trained for 30 epochs with seeds 40–44 and converted at $T \in \{4, 8, 16\}$. We compare MNE-L2,
 307 orthogonal weight regularization, no regularization, and optimizer L2. During the low-latency sweep,
 308 Gaussian noise is added after both the input IF and first hidden IF at $\sigma \in \{0, 0.25, 0.5\}$; each nonzero
 309 point averages three paired streams within each training seed.

310 For the compact CNN, the dense ANN baseline is 30,184 MACs per sample. The event counter
 311 charges every nonzero presynaptic activation once for each valid downstream connection. It counts
 312 both convolutional layers and the classifier, including edge effects from padding, and accumulates
 313 over all T timesteps. Reported means and standard deviations are across five independently trained
 314 checkpoints.

315 Figure 6 gives the compact low-latency view. At $T = 4$, the event-SynOps ratio ranges from 0.36 to
 316 0.45 on Fashion-MNIST and 0.63 to 0.65 on MNIST; at $T = 16$, it rises to 1.54–1.97 and 2.71–2.86.
 317 These are algorithmic operation counts rather than wall-clock latency or hardware energy [1, 26].

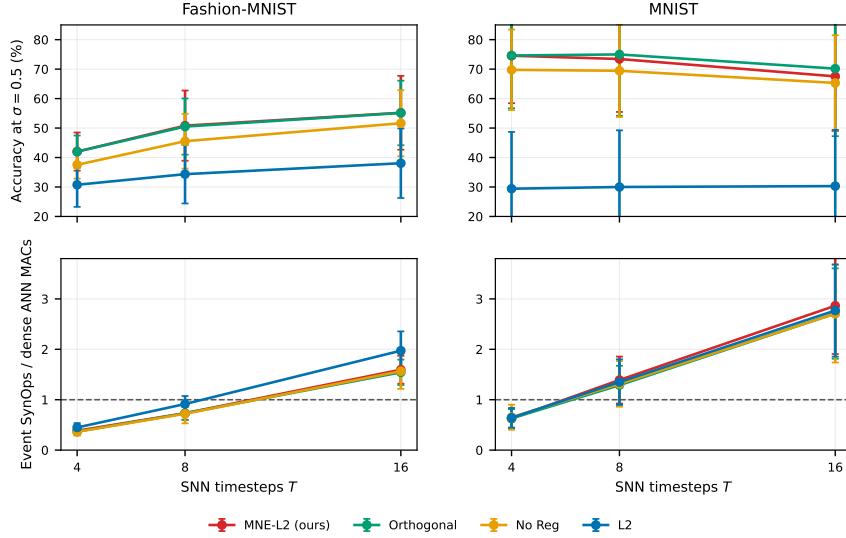


Figure 6: Appendix compact-model control. Top: five-seed accuracy under two-site Gaussian noise with $\sigma = 0.5$. Bottom: event-driven synaptic-operation proxy relative to one dense ANN MAC pass; the dashed line marks ratio 1. Error bars are standard deviations across training seeds.

Table 8: Full five-seed accuracy at $\sigma = 0.5$ (mean $\pm$ standard deviation across training seeds). Noise is injected after both the input IF and first hidden IF.

Dataset	Method	$T = 4$	$T = 8$	$T = 16$
Fashion	MNE-L2 (ours)	42.01 ± 6.48	50.81 ± 11.93	55.19 ± 12.53
Fashion	Orthogonal	41.98 ± 5.49	50.50 ± 9.54	55.12 ± 10.95
Fashion	No Reg	37.53 ± 4.69	45.53 ± 9.30	51.64 ± 11.19
Fashion	L2	30.75 ± 7.49	34.35 ± 9.95	38.06 ± 11.80
MNIST	MNE-L2 (ours)	74.55 ± 16.12	73.44 ± 17.98	67.54 ± 18.60
MNIST	Orthogonal	74.62 ± 17.89	74.99 ± 20.82	70.17 ± 22.92
MNIST	No Reg	69.75 ± 13.68	69.44 ± 15.69	65.31 ± 16.18
MNIST	L2	29.41 ± 19.27	30.00 ± 19.23	30.30 ± 19.07

D Reproducibility Notes

All reported datasets use their standard public train/test splits. The evaluation code fixes the training seed separately from the perturbation seed, stores every per-seed result before aggregation, and reports sample standard deviation rather than standard error. Figures 1, 3, and 5 are generated directly from stored CSV summaries by a deterministic plotting script; curve values are neither redrawn nor transcribed. The compact event-count control uses CPU inference and 1,024 examples per checkpoint. The CIFAR activity evaluation uses each full 10,000-example test set: sample mean, median, and p95 are computed within each seed, then summarized across seeds 40–44 by mean and sample standard deviation. Only $\sigma = 0$ activity enters energy; $\sigma = 1$ activity is retained as a diagnostic and excluded from deployment estimates. Full VGG and ResNet training requires GPU compute; evaluation does not update model parameters.

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657 Question: Does the paper describe the usage of LLMs if it is an important, original, or
658 non-standard component of the core methods in this research? Note that if the LLM is used
659 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
660 scientific rigor, or originality of the research, declaration is not required.

661 Answer: [N/A]

662 Justification: LLMs are not an important, original, or non-standard component of the
663 research method.

664 Guidelines:

- 665 • The answer [N/A] means that the core method development in this research does not
666 involve LLMs as any important, original, or non-standard components.
- 667 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
668 be described.